

A FULL LOW- p EXTENSION OF SPACE-TIME SHALLOW-NETWORK APPROXIMATION

AGENT LANE 14

STATUS

This packet upgrades the previous type- p partial note to a candidate full solution of the low- p future-work line in Abdeljawad–Dittrich [1]. The source theorem is stated for Bochner-Sobolev error exponents $2 \leq p_1 \leq p_2$, because the proof uses a type-2 Maurey estimate. The packet proves that the same $N^{-1/2}$ rate continues to hold in the entire low range

$$1 \leq p_1 \leq p_2 \leq 2.$$

In particular, the single-exponent case $1 \leq p < 2$ mentioned in the introduction is covered, including the endpoint $p = 1$.

SOURCE TARGET

The source explicitly marks the low- p case as future work:

At the present moment our analysis is limited to measuring the error in the Bochner-Sobolev norm with $p \geq 2$. This is because the currently available techniques and methods rely on the fact that the error is measured in a type-2 Banach space. For $p < 2$, the Bochner-Sobolev space will be a type- p Banach space and therefore, the existing theory does not apply anymore. We leave the investigation of this line of research for future projects.

In text form, the authors write that their analysis is limited to Bochner-Sobolev error with $p \geq 2$ because the available method uses a type-2 Banach space; for $p < 2$, the Bochner-Sobolev space is type- p , and they leave that line for future projects [1, p. 4].

PROOF INTUITION

The source proof already contains more structure than the type-2 explanation uses. In the dictionary-bound step, before taking any L^{p_1, p_2} norm, it proves a uniform pointwise bound for every relevant derivative of every scaled neuron. Therefore one does not need Banach-space type to sample the representing measure in the low- p range.

Sample the explicit signed measure representation of f that the source constructs. For each derivative and each point (x, y) , the sampling error is a scalar average of uniformly bounded independent random variables, so its p_2 -moment is $O(N^{-p_2/2})$ for $1 \leq p_2 \leq 2$. Since $p_1 \leq p_2$, Jensen's inequality lets this pointwise estimate pass through the mixed $L^{p_1}(U; L^{p_2}(V))$ norm. Summing over finitely many Sobolev derivatives gives the same $N^{-1/2}$ rate as the high- p theorem, now for all $1 \leq p_1 \leq p_2 \leq 2$.

A POINTWISE SAMPLING LEMMA

Lemma 1. *Let U, V be bounded measurable sets, let $1 \leq p_1 \leq p_2 \leq 2$, and let $n_1, n_2 \in \mathbb{N}$. Let $\mathbb{D}$ be a family of functions on $U \times V$ such that, for every $h \in \mathbb{D}$ and every pair of multiindices $|\alpha| \leq n_1$, $|\beta| \leq n_2$,*

$$|\partial_x^\alpha \partial_y^\beta h(x, y)| \leq H_{\alpha, \beta} \quad ((x, y) \in U \times V).$$

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Suppose

$$f = \int h d\mu(h)$$

in distributions, where μ is a finite signed Borel measure supported on $\mathbb{D}$. Put $M = \|\mu\|$. Then for every $N \geq 1$ there is an $f_N \in \Sigma_{N,M}(\mathbb{D})$ such that

$$\|f - f_N\|_{W_{n_1, p_1}^{n_2, p_2}(U, V)} \leq C_{\mathbf{n}, \mathbf{p}, H} M |U|^{1/p_1} |V|^{1/p_2} N^{-1/2}.$$

Proof. Write $d\mu = s d|\mu|$, with $|s| \leq 1$, and let $\rho = |\mu|/M$ if $M > 0$; the case $M = 0$ is trivial. Let $h_1, \dots, h_N$ be independent ρ -distributed samples and set

$$f_N = \frac{M}{N} \sum_{j=1}^N s(h_j) h_j.$$

Then $f_N \in \Sigma_{N,M}(\mathbb{D})$. Fix α, β and write

$$G_{\alpha, \beta} = \partial_x^\alpha \partial_y^\beta (f_N - f).$$

For each (x, y) , this is the mean of N centered scalar random variables, each bounded in modulus by $2MH_{\alpha, \beta}$. Hence, for $1 \leq p_2 \leq 2$,

$$\mathbb{E}|G_{\alpha, \beta}(x, y)|^{p_2} \leq (2MH_{\alpha, \beta})^{p_2} N^{-p_2/2}.$$

Using Fubini and Jensen's inequality with the concave exponent $p_1/p_2 \leq 1$,

$$\begin{aligned} \mathbb{E}\|G_{\alpha, \beta}\|_{L^{p_1}(U; L^{p_2}(V))}^{p_1} &= \mathbb{E} \int_U \left(\int_V |G_{\alpha, \beta}(x, y)|^{p_2} dy \right)^{p_1/p_2} dx \\ &\leq \int_U \left(\int_V \mathbb{E}|G_{\alpha, \beta}(x, y)|^{p_2} dy \right)^{p_1/p_2} dx \\ &\leq (2MH_{\alpha, \beta})^{p_1} |U| |V|^{p_1/p_2} N^{-p_1/2}. \end{aligned}$$

Therefore some sample satisfies the corresponding bound for all finitely many derivatives after summing the expectations over $|\alpha| \leq n_1$ and $|\beta| \leq n_2$. This is the asserted Sobolev estimate, with the finite derivative sum absorbed into $C_{\mathbf{n}, \mathbf{p}, H}$. $\square$

LOW- p THEOREM

Theorem 1. *Use the notation and the activation, weight, Fourier-Lebesgue, bounded-domain, and scaled-dictionary hypotheses from the proof of [1, Theorem 3.9]. Replace the source condition $2 \leq p_1 \leq p_2 < \infty$ by*

$$1 \leq p_1 \leq p_2 \leq 2.$$

Let $1 \leq q_i \leq 2$ and let $f \in \mathcal{FL}^{q_1, q_2}(\tilde{\omega})$. For M sufficiently large there is a constant $C > 0$, independent of N and f , such that

$$\inf_{f_N \in \Sigma_{N,M}(\mathbb{D}_{\tilde{\sigma}}^{d_1, d_2})} \|f - f_N\|_{W_{n_1, p_1}^{n_2, p_2}(U, V)} \leq C N^{-1/2} |U|^{1/p_1} |V|^{1/p_2} \|f\|_{\mathcal{FL}^{q_1, q_2}(\tilde{\omega})}$$

for every $N \geq 1$.

Proof. The source proof of Theorem 3.9 constructs an explicit signed measure representation of f by the scaled shallow dictionary $\mathbb{D}_{\tilde{\sigma}}^{d_1, d_2}$. Its variation estimate gives a measure μ with

$$\|\mu\| \leq A \|f\|_{\mathcal{FL}^{q_1, q_2}(\tilde{\omega})}$$

where A depends only on the structural parameters of the theorem. The same argument also proves that

$$f \in \mathcal{FL}^1(\omega(\cdot, \cdot) \langle \cdot \rangle \langle \cdot \rangle) \subseteq \mathcal{FL}^1(\omega).$$

By the source's low-degree embedding lemma [1, Lemma 3.7], applied with $t_1 = t_2 = 1$, this places f in $W_{n_1, p_1}^{n_2, p_2}(U, V)$ for all $1 \leq p_i \leq 2$.

It remains to approximate the representing measure. In the dictionary-bound part of the source proof, the authors show pointwise, before integration, that for every $|\alpha| \leq n_1$ and $|\beta| \leq n_2$,

$$|\partial_x^\alpha \partial_y^\beta \tilde{\sigma}(\cdot, \cdot; \xi, b)(x, y)| \leq H_{\alpha, \beta}$$

uniformly in (ξ, b) and $(x, y) \in U \times V$; this is the estimate that then yields their displayed L^{p_1, p_2} dictionary bound

$$\|\tilde{\sigma}(\cdot, \cdot; \xi, b)\|_{W_{n_1, p_1}^{n_2, p_2}(U, V)} \leq B|U|^{1/p_1}|V|^{1/p_2}.$$

Lemma 1, applied to this representing measure and these pointwise derivative bounds, gives an N -term network $f_N \in \Sigma_{N, \|\mu\|}(\mathbb{D}_{\tilde{\sigma}}^{d_1, d_2})$ with

$$\begin{aligned} \|f - f_N\|_{W_{n_1, p_1}^{n_2, p_2}(U, V)} &\leq C_{\mathbf{n}, \mathbf{p}, H} \|\mu\| |U|^{1/p_1} |V|^{1/p_2} N^{-1/2} \\ &\leq C N^{-1/2} |U|^{1/p_1} |V|^{1/p_2} \|f\|_{\mathcal{D}^{q_1, q_2}(\tilde{\omega})}. \end{aligned}$$

Taking $M \geq \|\mu\|$ proves the asserted infimum bound. $\square$

VERIFICATION AND SCOPE

The proof is symbolic; no numerical verification is needed. The source crop was generated by `code/make_open_problem_crop.py`. Human review should check the passage from the source's pointwise derivative estimate to Lemma 1; this is the entire upgrade mechanism.

This packet fully covers the source's stated low- p obstruction for the same two-block shallow-network theorem, including the single-block case $p_1 = p_2 = p < 2$ and the endpoint $p = 1$. It does not address separate future directions in the paper such as Bochner-Besov targets or arbitrary Bochner-Banach spaces.

NOVELTY CHECK

The local lightweight indexes were searched for 2312.08461, the paper title, **Fourier Lebesgue**, **Bochner-Sobolev**, **type-p**, $p < 2$, and neural-network approximation keywords. The only existing packet was the earlier lane-14 partial note, which this packet replaces and strengthens.

Web searches on 2026-07-18 for the arXiv id, the exact title, and phrases combining low- p or $p < 2$ Bochner-Sobolev error, shallow neural networks, Fourier-Lebesgue spaces, and the rate $N^{-1/2}$ found the source paper and generic related material, but no later resolution of this low- p extension.

REFERENCES

- [1] A. Abdeljawad and T. Dittrich, *Space-Time Approximation with Shallow Neural Networks in Fourier Lebesgue spaces*, arXiv:2312.08461, 2023.